

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Shipped real product changes end to end on a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via REST integrations; contributed to **\$340K+** in revenue.
- Built a full-stack platform (**React**, Node.js, MongoDB) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Piloted a COO-sponsored GitHub Copilot experiment and used AI-assisted development to ship faster, hardened auth, added AWS KMS encryption, and remediated **20+ VAPT findings**.
- Served as primary technical POC, fixing customer-reported production bugs by debugging incidents to root cause and owning fixes across the stack.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) returning JSON, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production feature running a **4-bit quantized Llama-7B** model via llama.cpp with async queued generation.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN (**React** front end) platform with REST APIs, a multi-branch data model, and cron automation, built and shipped independently end to end; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage **agentic research** service in **Python** that turns an open-ended prompt into a structured entity table via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

cf-ai-research-scout – Agentic Research Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built a full-stack AI research assistant with durable, multi-step agent workflows that coordinate tool calls, survive process restarts and tool failures, and stream intermediate results to the client over WebSocket in real time.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a knowledge-graph-guided RAG pipeline (query decomposition, RRF fusion, cross-encoder reranking, MMR diversification), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Java, Bash

Frontend & Full-Stack: React, Next.js, Node.js, Express, FastAPI, REST APIs

AI & Agents: LLM agents, RAG, PyTorch, Hugging Face Transformers, sentence-transformers, MCP, evaluation

Data & Tools: pandas, NumPy, Spark, Docker, Git, GitHub Actions, PostgreSQL, MongoDB, SQLite

Cloud / Platforms: AWS, GCP, Azure, Linux, Cloudflare Workers, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.